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Effects of Recommendation Neutrality and Sponsorship Disclosure on Trust vs. Distrust in Online Recommendation Agents: Moderating Role of Explanations for Organic Recommendations

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Abstract. We extend the extant research on neutral recommendation agents (RAs) to those that lack recommendation neutrality and are biased toward sponsors. We first investigate the effects of recommendation neutrality on users' trust and distrust in RAs by comparing a biased RA with sponsorship disclosure with a neutral RA. We then apply a contingency approach to examine the effects of sponsorship disclosure on users' trust and distrust in *biased* RAs, with explanations for organic recommendations as a contingent factor. A laboratory experiment was conducted in the United States. We determine that users' trust in the biased RA with sponsorship disclosure is lower and that their distrust is higher than that in the neutral RA. Results also show that user *trust* in a biased RA increases only when explanations for organic recommendations and sponsorship disclosure are both provided. Users' perceived psychological contract violations of an RA have been verified as a key mediator of the examined effects. However, explanations for organic recommendations, sponsorship disclosure, or their combination fail to significantly lower users' *distrust* in a biased RA. A second experiment conducted in Hong Kong confirms the major findings of the experiment conducted in the United States. Theoretical contributions and practical implications for e-commerce RAs are discussed.

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1. Introduction

Most existing research on online recommendation agents (RAs) focuses on *neutral* RAs designed to guide online consumers in product selection (Dellaert and Häubl 2012, Song et al. 2007, Wang and Benbasat 2013, Xu et al. 2014). These neutral RAs use their search engines to identify and recommend products to users. However, online sponsorships are widely adopted in search engine–related technologies, including RAs, to promote the products of technology sponsors (e.g., Cheng et al. 2011, Harvey 2003, Ransom 2010, Yang and Ghose 2010). Neutral RAs recommend products solely based on users' requirements and sort these products according to the suitability of users' requirements; by contrast, biased RAs can manipulate the recommendation order to favor sponsors' products at the top of the recommendation page regardless of the relevance of these sponsored products to users' requirements (Grimmelmann 2010). Many RAs are no longer neutral in their recommendations; instead, they are biased toward sponsored products that may not exactly fit users' requirements. For example, a sponsored product supposed to be listed in the seventh position (in terms of fitting user requirements) of the recommended products by neutral RAs will be moved to the top position by a biased RA with the original top six products moved down in the list. Users can be exploited by a biased RA, and may consider the provision of biased recommendations as an injurious conduct (Xiao and Benbasat 2011).

Users' trust and distrust are a critical concern among *biased* RAs (Estelami 2016). Previous research confirms that trust is a key factor that affects users' acceptance of a *neutral* RA (e.g., Wang and Benbasat 2005), but this research line has generally focused on *trust*. This condition is understandable because *distrust* is less relevant for neutral RAs. However, users' distrust in a biased RA can be triggered when they are aware of recommendation biases (i.e., lack of recommendation neutrality). The current research extends previous research on neutral RAs to *biased RAs* and focuses on both *trust* and *distrust* in RAs.

Trust and distrust are not simply the opposite ends of a single continuum (Dimoka 2010, Komiak and Benbasat 2008, Lewicki et al. 1998, McKnight and Choudhury 2006, Sitkin and Roth 1993). Trust is a positive-valent belief and an expectation of the beneficial conduct of others. By contrast, distrust is a negative-valent belief and an expectation of others' injurious conduct. The low expectation of others' beneficial conduct does *not* necessarily signify high chances of obtaining the injurious conduct of others. Therefore, trust and distrust are separate, albeit closely associated, constructs. For example, users may not trust an RA if they do not believe that it can determine the most suitable products for them. However, this low level of trust does not necessarily imply that users distrust the RA and consider that the RA would take advantage of them (e.g., by recommending biased products). Previous research suggests that "there are elements that contribute to the growth and decline of trust, and there are elements that contribute to the growth and decline of distrust" (Lewicki et al. 1998, p. 440).

To mitigate the potential harmful effects of a biased RA on trust and distrust, some companies clearly disclose involved sponsorships. However, other companies fail to disclose such information unintentionally or deliberately. The question is when biased RAs disclose the involved sponsorships, whether this disclosure will result in the same level of high trust and low distrust as neutral RAs. In addition, can sponsorship disclosure make a difference in increasing users' trust and lowering their distrust in biased RAs? These research questions merit examination considering that not all biased RAs provide sponsorship disclosure to their users. Consumer Reports *WebWatch* noted that many technology providers "seem to do as little as possible to explain their relationship with advertisers" (Fallows 2005). Danny Sullivan, a widely cited authority on search engines, noted in June 2012 that many search engines (including Google) and related RA technologies fail to disclose sponsorships clearly.[1] For example, Google Flight Search vaguely states that "Google may be compensated by some of these providers" when returning a list of major airlines' price quotes for an air-ticket search. In addition, as reported by the *New York Times*, Google's search engine mysteriously put JCPenney in the first spot for dozens of dresses and bedding-related terms without clearly disclosing the sponsorship involved (Segal 2011). The prevalence of biased RAs without clear sponsorship disclosure indicates that many such platforms assume that the disclosure of biased RAs would result in users' *negative* responses (Fallows 2005, Guijarro et al. 2015). Hence, examining how such sponsorship disclosure, and under which circumstances, really influences users' trust and distrust in biased RAs is particularly important. Our research results can provide implications for RA designers on whether biased RAs should disclose sponsorship clearly and whether websites providing neutral RAs should include sponsored products in their recommendation for improved profit without damaging users' trust and/or raising their distrust.

To address the above questions, the first objective of this study is to examine the difference between a biased RA with sponsorship disclosure and a neutral RA—i.e., the effect of recommendation neutrality. We draw on the perspective of psychological contract violations (PCV) (Pavlou and Gefen 2005, Robinson 1996) to compare biased RAs with sponsorship disclosure and neutral RAs with respect to users' trust and distrust. Psychological contracts are crucial to the understanding of buyer–seller relationships in noisy environments (e.g., online markets), and the PCV perspective properly explains the buyer transaction behavior in e-marketplaces (Pavlou and Gefen 2005). Online consumers establish psychological contracts to guide their trust relationships and interactions with others. Although recent studies have investigated the sponsorship phenomena in online environments, they mainly focus on the effectiveness of such sponsorships from the sponsor's perspective (i.e., benefits for the sponsors) (Ghose and Yang 2009, Rutz et al. 2011). This literature gap provides an opportunity to study the RA neutrality phenomenon from the users' perspective, particularly from the PCV perspective.

After comparing a biased RA with a neutral RA, the question that naturally follows is how to significantly improve users' trust in biased RAs. Will the provision of explanations for its organic recommendations by a biased RA with sponsorship disclosure help? This study addresses these questions as its *second objective*. That is, we aim to investigate the effects of the information provided by a biased RA (including sponsorship disclosure) on users' trust and distrust in the biased RA. Previous research on RAs investigates the effects of explanations provided by a *neutral* RA on users' trust on the neutral RA (Wang and Benbasat 2007). These explanations can be applied to biased RAs as well. Thus, a biased RA can not only disclose the sponsorships involved, but also explain the inner workings of its recommendations that are solely based on users' requirements (these explanations are referred to as explanations for organic recommendations or explanations in brief hereafter). What are the roles of these sponsorship disclosures and explanations with respect to users' trust and distrust in a *biased* RA? Will sponsorship disclosure lead to positive outcomes regardless of the provision of the explanations? Can a biased RA selectively provide these explanations, or sponsorship disclosure, or both? These are important questions given that *not* all biased RAs explain how their organic recommendations are generated (e.g., Expedia.com and Epinion.com) or provide sponsorship disclosure (e.g., Apartmentguide.com; see Edelman and Gilchrist 2012).

By achieving the two objectives identified above, we expect to make the following contributions. First, by studying the effects of recommendation neutrality and sponsorship disclosure from the users' perspective (i.e., the PCV perspective), our study reveals the underlying mechanism that enables us to extend the theory on how trust and distrust are affected. Second, we contribute to the RA literature by investigating the different effects of RA designs on users' trust *versus* distrust. This investigation is important because prior RA research predominantly focuses on neutral RAs and users' trust in these RAs (Wang and Benbasat 2008). Few studies have examined biased RAs, in which distrust is a more salient issue. The present study demonstrates that although the faithful provision of both explanations and disclosure of sponsorships can mitigate the negative effects of the biased recommendations on trust, such provision cannot resolve users' *distrust* in a biased RA. Third, we extend the theoretical boundaries by examining how the explanations provided by RAs can (or cannot) moderate the effects of sponsorship disclosure on trust and distrust. Thus, we also contribute to literature by investigating how and when sponsorship disclosure affects trust and distrust. Existing literature has yet to clarify whether biased RAs' sponsorship disclosure always has a positive effect on trust, which explains why some firms have failed to provide clear sponsorship disclosure. Our findings can inform RA practitioners on whether providing biased RAs is a suitable idea and whether disclosing sponsorship in the presence of explanations can help mitigate the negative effects that result from providing biased recommendations.

2. Background

2.1. Neutral and Biased RAs

This study was conducted in the context of RAs using a method based on content filtering.[2] Previous studies on this type of RA predominantly focus on neutral RAs (e.g., Song et al. 2007, Wang and Benbasat 2009, Xu et al. 2014). Neutral RAs first explicitly elicit the user's preferences on product attributes and then generate the personalized product advice that is most suitable for the user by faithfully following a decision strategy to process the elicited preferences. These RAs can be considered sophisticated forms of search engines offered by many individual e-commerce websites (e.g., BestBuy.com (Lee and Benbasat 2011) and Sears.com (Murray et al. 2010)), as well as by aggregators (e.g., Priceline.com), given that they all obtain users' preferences and then provide unbiased product recommendations. For example, BestBuy's "Find the Perfect TV" RA provides an unbiased list of products that fit users' indicated preferences. When a user specifies a TV size lower than 40 inches and TVs with smart features, it will recommend TVs that faithfully fulfill the user's requirements without listing any sponsored products. Similarly, if you search for "Samsung TV," then BestBuy.com will generate a list of Samsung TVs only.

Many commercial RAs are *biased* toward sponsors because of the wide use of sponsorships in search engine–related technologies (Guijarro et al. 2015, Ransom 2010, Yao and Mela 2011). Online sponsorships are the primary business model for many search engines and related RA technologies (Goo 2007, Katona and Sarvary 2010). Online sponsorships for RAs (and search engine–related technologies in general) are frequently implemented with a pay-for-placement model in the recommendation results (Goo 2007, Yao and Mela 2011). Potential sponsors bid for the right to present a sponsored product and list the sponsored product before others on the search results page (or product recommendation page) in this model (Dhar and Ghose 2010).

In this study, biased RAs recommend products that do not fit users' indicated preferences, which is consistent with prior literature's definition of biased RAs (Komiak and Benbasat 2006, Krishnasamy et al. 2015, Grimmelmann 2010, Xiao and Benbasat 2015). We focus on this type of biased RA mainly because the chances that such biases can be detected are high. For example, if a user strongly prefers a product with a price less than $500, and the recommended products still include products priced higher than $500 (other things being equal), then it will most likely be detected by the user. Xiao and Benbasat (2015) manipulated a biased RA by recommending less desirable products (with two product attributes dominated by a better product) with the same brand name as the user specified. They showed that users perceived a biased RA as more biased than an unbiased RA even when sponsorship disclosure is absent, which indicates that users can realize that they are using a biased RA.

This study examines a situation in which biased recommendations are perceptible to users. If a user cannot discern a biased RA, then his/her trust and distrust in the RA will remain unaffected by the biased recommendations. An RA can also be unnoticeably biased by considering other factors when determining its recommendation order than those related to users' indicated preferences. For example, an RA can recommend a product set that fits user preferences but sorts the products in a certain order that favors the product with the highest price or newly released products. These bias types are outside of the scope of the current study because sponsorships are not necessarily involved in these biases.[3]

Biased RAs exist in many different industries. Apartment websites (e.g., Homefinder.com) organize recommendation results by advertiser payment (Edelman and Gilchrist 2012). Amazon.com provides a biased list of products that may not necessarily all fit the user's specified requirements. Unlike in the case of

BestBuy mentioned earlier, if you search for "Samsung TV," Amazon.com lists not only Samsung TVs, but also sponsored products (e.g., TCL and Sony TVs) on the first page of the search results. Similarly, Expedia.com and Epinions.com frequently list sponsored products at the top of the search and recommendation results pages. However, some biased RAs fail to clearly disclose the bias and sponsorships involved,[4] probably because of concerns about potential negative responses of users once users are informed of the use of sponsorship practices (Fallows 2005). Examples include TripAdvisor (Sullivan 2002), Apartmentguide.com (Edelman and Gilchrist 2012), pharmaceutical websites (Liang and Mackey 2011), and some health and financial websites (Stanford et al. 2002). The absence of sponsorship disclosure is even more prevalent in unbranded websites (Liang and Mackey 2011) and in countries that lack regulations regarding Internet search and recommendation results.

RAs use search engines to recommend products, but they differ from search engines in two main aspects. First, search engines search items at a broader level (i.e., unrestricted to products available in a particular website), whereas RAs used in e-commerce search for products available only in a particular website. Second, search engines provide broader services (e.g., users can ask a general question, such as, "What is a smart phone?"), yet RAs commonly allow a user to specify particular product requirements or infer the user's preferences. This study focuses on RAs to enhance internal control in the conducted laboratory experiments. However, we expect that our findings can be extended to search engines when users employ a search engine to find a specific product to buy. In this case, a search engine serves the purposes of an RA because (1) the RA uses a search engine embedded to identify products suitable for the user and (2) both are supposed to return output by considering users' requirements and preferences.

2.2. Trust and Distrust in RAs

In the RA context, trust is widely conceptualized as a positive belief in an RA. Three widely acknowledged trusting beliefs are competence, integrity, and benevolence (McKnight et al. 2002; Wang and Benbasat 2007, 2016). The competence belief refers to a user's perception that an RA has the skills and expertise to perform effectively in specific domains. The integrity belief is the perception that the RA is designed to adhere to a set of principles (e.g., honesty and keeping promises) generally accepted by users. The benevolence belief refers to the user's perception that the RA is designed to consider and act in the user's interest. These beliefs reflect users' implicit assumptions when interacting with an a priori trustworthy RA given that the recommendation services are provided by the website to facilitate their online decision making. These assumptions are largely taken for granted and unacknowledged until violated (Luhmann 1979, Robinson 1996, Zucker 1986).

By contrast, distrust is a negative-valent belief in an RA. Corresponding to the three trusting beliefs, the distrusting belief also includes three dimensions related to the competence, integrity, and benevolence of an RA. Recent literature also adopts a two-dimensional view of distrust that combines the first two beliefs (related to competence and integrity) under the umbrella of discredibility, thereby turning discredibility and malevolence into two major distrusting beliefs (Dimoka 2010). Discredibility deals with the trustor's concerns regarding the trustee's dishonesty and competence in engaging in harmful behavior, indicating that the trustee can act deceptively and irresponsibly. Malevolence relates to concerns regarding the trustee's conduct in terms of a malicious intention that can hurt the trustor's welfare. A malevolent trustee can exploit the trustor's vulnerability.

As a negative-valent belief, distrust is rooted in goal and value incongruence between the trustee and trustor (Connelly et al. 2012, Hardin 2004, Lewicki et al. 1998). Such incongruence portends the general suspicion that the trustee can behave opportunistically to exploit the trustor. Moreover, goal and value incongruence is persistent and is difficult to resolve once distrust is formed (Connelly et al. 2012, Dirks and Skarlicki 2004).

Recent research advances in the area of trust and distrust confirm that these two constructs are qualitatively different from each other (Benamati et al. 2006, Connelly et al. 2012, Dimoka 2010, Komiak and Benbasat 2008, Ou and Sia 2010). Distrust is not merely the absence of trust; trust and distrust can coexist in an individual's cognitions. The foundational support for the separation of trust and distrust comes from the early works of Luhmann (1979) and Kahneman and Tversky (1979), who distinguished positive and negative valence constructs and laid the groundwork to theorize that trust and distrust are distinct constructs. Cacioppo and Berntson (1994) and Cacioppo et al. (1997) then reviewed evidence that the positive and negative evaluative processes underlying different attitude-related constructs are distinguishable with distinct antecedents and consequences.

Lewicki et al. (1998) further elaborated the multifaceted and multiplex nature of relationships and argued that consumer ambivalence (e.g., simultaneous trust and distrust) is common in a relationship-building process. People can simultaneously trust and distrust one another given their varying experiences related to the different relationship facets. Cognitive neuroscience studies that use functional neuroimaging tools confirm that trust and distrust span distinct brain areas and are distinct constructs associated with different neurological processes (Dimoka 2010). In the RA context, Komiak and Benbasat (2008)

Table 1. Coexistence of Trust and Distrust in the RA Context

	Low distrust	*High distrust*
	Characterized by • No fear • Low vigilance	Characterized by • Fear • High vigilance
High trust Characterized by • Faith • Initiative	• Belief in an RA's recommendations that can fit their requirements • Reliance on an RA's recommendations for product selection • No fear that an RA's recommendations can harm them • No vigilance during interaction with an RA	• Belief in receiving recommendations from an RA that can fit users' requirements • Reliance on an RA's recommendations for product selection, but also high vigilance during interactions with the RA • Fear in an RA's recommendations that can harm them and monitor for potential vulnerabilities
Low trust Characterized by • No faith • Passivity	• Lack of belief that an RA's recommendations can fit their requirements, but also no fear that the RA's recommendations can harm them • No vigilance during interaction with an RA • No initiative to use an RA actively	• Lack of belief that an RA's recommendations can fit their requirements • Fear that an RA's recommendations can harm them, and monitor for potential vulnerabilities • High vigilance during interaction with an RA • No initiative to use an RA actively

adopted concurrent verbal protocols from 49 subjects by using RAs and revealed that trust-building and distrust-building processes are distinct and separate. The distinct and separate trust and distrust constructs are also evident in prior studies involving banking (Benamati et al. 2006), auction (Dimoka 2010), and general online users (Cho 2006, McKnight and Choudhury 2006, Ou and Sia 2010).

Adapting the four prototypical relationship conditions about trust and distrust of Lewicki et al. (1998) to the RA context, Table 1 shows the coexistence of trust and distrust at different levels. Users believe that an RA's recommendations fit their requirements and do not question the RA's behavior in the top left (high trust/low distrust). Thus, users will initiate the use of RA and rely on its recommendations to select products. By contrast, users have no reason to believe the RA's recommendations in the lower right (low trust/high distrust). These users also fear that the RA can engage in harmful behavior by pursuing its own interest.

Users believe an RA's recommendations (e.g., its organic recommendations) in the top right (high trust/high distrust), but they also have reasons to be extremely wary of the RA's recommendations (e.g., its sponsored recommendations). For example, organic recommendations on Expedia.com are generated based on users' requirements, which establish the foundation for users' trust in them. However, Expedia also works in their sponsors' interests. The provision of sponsored recommendations can be considered a form of user exploitation because following these recommendations can hurt the users' interests. Thus, users also maintain vigilance toward the recommendation results and monitor whether these results fit their requirements. Many users of RAs (and search engine–related technologies in general) have been reported to trust these technologies to provide relevant information and trustworthy recommendations; however, they are also distrustful and concerned about potential exploitation because of the profit-seeking nature of these technologies (Winder 2011).

Users have no reason to believe or be wary of an RA's recommendations in the bottom left (low trust/low distrust). These users do not heavily rely on the RA's recommendations in their product selection because of their low trust in the RA, but they may employ it casually given their low distrust of the RA.

3. Hypothesis Development

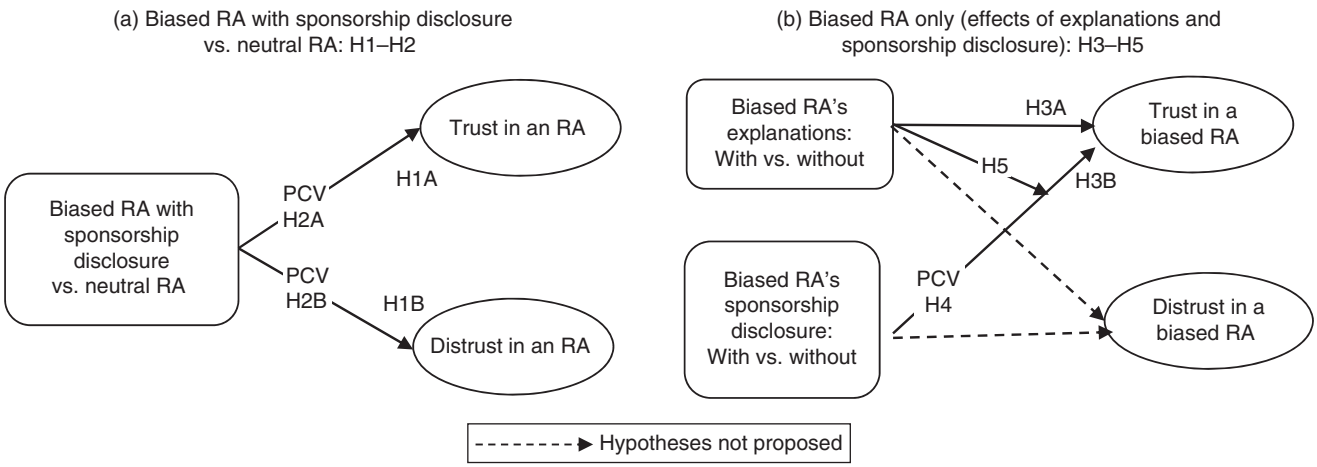
We first compare biased RAs with sponsorship disclosure and *neutral* RAs in terms of trust and distrust (Figure 1(a) and Section 3.1). Then, we focus on *biased RAs* only. We examine the effects of sponsorship disclosure and explanations for organic recommendations on trust and distrust in the biased RAs, and the contingent role of these explanations for the effects of sponsorship disclosure (Figure 1(b) and Sections 3.2 and 3.3).

3.1. Comparisons of Biased RAs with Sponsorship Disclosure and Neutral RAs

The effects of *recommendation neutrality* refer to the differences between neutral RAs and RAs *with* sponsorship disclosure. We exclude biased RAs *without* sponsorship disclosure from the comparisons with neutral RAs at this point since they are hypothesized in H3B to be worse than the biased RAs *with* sponsorship disclosure in terms of trust.

Extensive RA literature indicates that users generally expect an RA to provide personalized recommendations to them (see Xiao and Benbasat 2014 for a review). Personalized recommendations require an RA to understand and represent users' needs and preferences (Häubl and Murray 2006, Xu et al. 2014).

Figure 1. Summary of Hypotheses



In particular, an RA that is based on content filtering explicitly elicits users' needs and requirements (i.e., users consider the RA their agent and explicitly delegate some decision tasks to it). Therefore, users expect the RA to recommend products based on their indicated needs and preferences (Dou et al. 2010, Nicholson et al. 2006, Wang and Benbasat 2007). Users may believe that the RA is obligated to perform a certain behavior (e.g., provide product recommendations that fit their indicated needs and preferences) based on this assumed agency relationship and explicit elicitation of user requirements. This scenario constitutes a psychological contract between an RA and its users (Pavlou and Gefen 2005, Robinson 1996, Rousseau 1995).

A psychological contract is essential in relationship building in the e-marketplace context (Pavlou and Gefen 2005, Robinson 1996). Prior studies confirm that the violations of psychological contracts lead to lower trust in intraorganizational (Robinson 1996), interorganizational (Kingshott and Pecotich 2007), and e-marketplace contexts (Pavlou and Gefen 2005). These violations by other entities challenge the foundation of a person's general beliefs regarding others' codes of conduct associated with trusting relationships; therefore, trust in others will be shattered (Robinson 1996, Rousseau 1995).

A biased RA will normally place sponsored products as the top recommendations regardless of their relevance to users' preferences. However, users generally expect the most relevant products to be listed at the top of the recommendation results because they generally consider the rankings of the recommendations as indicative of their relevance to their needs and preferences (Dou et al. 2010, Kramer 2007). Therefore, the sponsored recommendations from a biased RA will deviate from the users' expectations, leading to the perceived PCV. For example, if a user prefers the Nikon brand along with a set of other preferences when searching for a digital camera, then a biased RA (e.g., the RA is sponsored by the digital camera manufacturer XYZ) can recommend XYZ cameras as its top recommendations. These XYZ cameras clearly fail to fit the user's requirement of Nikon cameras. Therefore, the user may believe that the biased RA violated the psychological contract.

When compared to the neutral RAs, the biased RAs even with sponsorship disclosure can still lead to a certain degree of PCV because not all recommendations provided by these biased RAs fit users' indicated preferences. Nevertheless, the PCV of a biased RA with sponsorship disclosure will not be as severe as that of a biased RA *without* sponsorship disclosure, as we will elaborate further for H3B (i.e., effects of sponsorship disclosure of biased RAs). Thus, PCV falls into a continuum ranging from low (a neutral RA) to high (a biased RA without disclosure), in which a biased RA with sponsorship disclosure falls between the two.

In the context of this study, when users perceive an RA's violations of psychological contracts because of its lack of neutrality, they are less inclined to believe that the RA will engage in beneficial conduct for them to fulfill their positive expectations. Although sponsorship is disclosed, it cannot completely eliminate users' long-held belief that RAs are supposed to be neutral (i.e., a psychological contract). Therefore, we propose the following hypotheses:

Hypothesis 1A (H1A). *Users' trust in a biased RA* with *sponsorship disclosure will be lower than that in a neutral RA* (*i.e., effect of recommendation neutrality*).

Hypothesis 2A (H2A). *The effect of recommendation neutrality on trust will be mediated by perceived PCV.*

We expect that user distrust in a biased RA with sponsorship disclosure will be higher than that in a neutral RA. The violations of psychological contracts due to biased recommendations create a sense of betrayal and

unfair treatment (Niehoff and Paul 2001, Pavlou and Gefen 2005), thereby leading to users' negative expectations of a biased RA. These violations can trigger users' suspicions that the RA has ulterior motives to seek profits from sponsors, and that it engages in opportunistic behavior that can lead to adverse outcomes (Fein 1996, Kramer et al. 1996, Lewicki et al. 1998). The high distrust in a biased RA cannot be easily reversed even with the sponsorship disclosure. By contrast, perceived PCVs have a minimal chance to exist in the case of neutral RAs. Thus, the level of distrust in neutral RAs will be lower. We then propose the following:

Hypothesis 1B (H1B). *Users' distrust in a biased RA* with *sponsorship disclosure will be higher than that in a neutral RA.*

Hypothesis 2B (H2B). *The effect of recommendation neutrality on distrust will be mediated by perceived PCV.*

3.2. Effects of Biased RAs' Explanations and Sponsorship Disclosure on Trust

To the best of our knowledge, prior studies on the effects of explanations provided by computer systems were generally conducted in the context of systems that work solely for users, such as neutral RAs. In particular, Wang and Benbasat (2007) investigated the *direct* effects of explanations on users' trust in neutral RAs. They found that in line with theories on knowledge-based trust development (Lewicki and Bunker 1995, McKnight et al. 1998), three explanation types can promote users' enhanced understanding of an RA. These types include *how* explanations, *why* explanations, and *guidance*. The *how* and *why* explanations inform users of system actions (e.g., how the RA works and why its actions are appropriate) and assist users in creating a valid mental model of an RA (Mcsherry 2005). *Guidance* provides information on the product attribute preference options to help users express their preferences when using the RA. The RA becomes transparent with these explanations and its behavior becomes increasingly predictable. Hence, users' trust in an RA can be improved when explanations are provided by the RA.

As discussed earlier, the difference between a neutral and biased RA is that the latter adjusts the product recommendation order by favoring the sponsored products. Nevertheless, biased RAs still recommend products that are supposed to be recommended by neutral RAs. When explanations for its organic recommendations are provided by a biased RA, users can recognize that not all recommendations are biased and that many recommendations are generated according to users' preferences. Users will understand the extent to which their preferences are considered by the biased RA, and this understanding can lead to user trust (Wang and Benbasat 2008). By contrast, for a biased RA *without* explaining its organic recommendations, users are uncertain of the extent to which such RA recommends products that fit their preferences. Thus, they will have lower trust in the biased RA. We then propose the following:

Hypothesis 3A (H3A). *Users' trust in a biased RA* with *explanations for its organic recommendations will be higher than that in a biased RA* without *these explanations.*

We then examine the effects of sponsorship disclosure on trust. In our context, psychological contracts entail beliefs and expectations that RAs should behave in a certain manner based on some implicit assumptions commonly conceived unilaterally by users (Robinson 1996). However, when a biased RA discloses the use of a sponsorship strategy, users' implicit assumption that the RA works solely for them is not completely valid. The users may have to adjust their expectations slightly regarding what they can receive from the RA and alter their psychological contract with the RA. Users may acknowledge that sponsored recommendations are natural results provided by a biased RA. Therefore, users will understand why sponsored products are presented and can consider the provision of sponsored recommendations as a violation of psychological contracts *to a lesser degree*. In line with knowledge-based trust theory (Lewicki and Bunker 1995, McKnight et al. 1998), sponsorship disclosure can lead to users' trust in a biased RA through lowered perceptions of PCV. Thus, we propose the following:

Hypothesis 3B (H3B). *Users' trust in a biased RA* with *sponsorship disclosure will be higher than that in a biased RA* without *sponsorship disclosure.*

Hypothesis 4 (H4). *The effect of a biased RA's sponsorship disclosure on trust will be mediated by perceived PCV.*

H3B posits the positive effect of a biased RA's sponsorship disclosure on trust in the biased RA. Nevertheless, sponsorship disclosure can only partially justify the recommendations of the biased RA (i.e., sponsored recommendations). The biased RA can provide organic recommendations as well. Should the biased RA fully justify all of its recommendations by providing both sponsorship disclosure and explanations for its organic recommendations?

Next, we examine the effects of sponsorship disclosure under two scenarios. In the first scenario, a biased RA does *not* provide explanations for its organic recommendations. Under this scenario, when a biased RA only discloses the sponsorships involved but does not provide these explanations, this disclosure may lead to mixed outcomes. On one hand, this disclosure can gain users' trust given that the biased RA is honest to the user. On the other hand, the user may suspect that many products recommended can be biased toward the sponsors. Thus, the user will perceive that the RA violates the psychological contract to a certain extent. Considering these two different possible

consequences, we contend that the positive effect of the sponsorship disclosure on trust in a biased RA will be limited when explanations are *absent*.

By contrast, under the second scenario where a biased RA *does* explain its organic recommendations, users can distinguish sponsored recommendations from organic ones, when the sponsorships involved are also disclosed. Users have a complete picture of how the biased RA works and can predict future RA performance. Users' perception of PCV is maintained at the minimum. Thus, trust in a biased RA can be significantly improved with the simultaneous provision of the explanations and sponsorship disclosure according to knowledge-based trust theory (Lewicki and Bunker 1995, McKnight et al. 1998). However, when a biased RA only provides explanations for its organic recommendations but does *not* disclose the sponsorships involved, then the explanations can mislead the users to believe that all products recommended by the biased RA fit the users' need, which will eventually lose the users' trust. In other words, the explanations will be most needed and effective in gaining users' trust in a biased RA that provides sponsorship disclosure. These explanations for organic recommendations can magnify the positive effects of the sponsorship disclosure of biased RAs on trust. Thus, we propose the following:

Hypothesis 5 (H5). *The positive effect of a biased RA's sponsorship disclosure on trust will be contingent on its explanations for organic recommendations. The effect will be* larger *when the biased RA provides these explanations than when it does not* (*i.e., a positive moderating effect of the explanations*).

3.3. Effects of Biased RAs' Explanations and Sponsorship Disclosure on Distrust

As widely acknowledged in the literature, trust and distrust are distinct constructs; thus, the effects of explanations and sponsorship disclosure on distrust can vary from those on trust. Distrust is rooted in goal and value incongruence and cannot be resolved easily (Connelly et al. 2012). Table 1 shows that high distrust is characterized by fear and vigilance because users are concerned about an RA's possible engagement in harmful behavior to exploit their vulnerability. Furthermore, distrust is more emotionally charged than trust (McKnight and Chervany 2001). The negative emotions associated with distrust, such as doubt, worry, panic, and anger, cause the distrustor to adopt a wary and defense-minded posture (McKnight et al. 2004, Wang et al. 2016). Such a posture largely tends to be challengingly mitigated by explanations or sponsorship disclosure provided by a biased RA.

Therefore, unlike trust, which can be improved from the state of lack of belief to a great faith when an RA provides explanations and sponsorship disclosure, users' fear of a biased RA cannot be easily removed through explanations or sponsorship disclosure. Instead, the explanations and disclosure may compel the users to become vigilant and wary of the potentially opportunistic behavior of the biased RA. This scenario is due to the biased RA's explanations and sponsorship disclosure increasing the accessibility of its ulterior persuasive motive (Campbell and Kirmani 2000), which is to persuade users to select the sponsored products. Explanations for organic recommendations cannot explain the recommended biased products, but they can instead raise suspicion that the biased RA attempts to hide its true motives in recommending these products. Sponsorship disclosure confirms users' thought that the RA is biased, thereby activating users' perceived ulterior motives of a biased RA. Users can further realize that sponsorship disclosure is a tactic of the biased RA to gain users' trust. The marketing literature on persuasion knowledge indicates that the accessibility of a sales agent's ulterior motives influences users' negative perceptions of that agent (Campbell and Kirmani 2000). For example, if consumers infer that a salesperson's ulterior motive is to sell products, then they will consider the motive of that salesperson's flattering remark (even though it can be true) as insincere.

The accessibility of an ulterior motive depends on users' perception of the agent's motive and its recency of activation (Campbell and Kirmani 2000, Higgins and King 1981). Users may consider the provision of sponsored recommendations as an ulterior motive of the RA in our context. The disclosure of sponsorship and explanations (recency of activation) of a biased RA may increase the accessibility of the RA's ulterior persuasive motive. Although the RAs can be honest, the heightened accessibility of the biased RAs' ulterior motives can create general suspicions and distrust (Campbell and Kirmani 2000).

However, the explanations and sponsorship disclosure have benefits, as argued in H3A and H3B. They can increase users' understanding of a biased RA's behavior. Users can also adjust their expectations of what they can receive from the biased RA; thus, the negative effect of the explanations and sponsorship disclosure on distrust can be lowered. In other words, their benefits of improving trust also extend to lower distrust, which is illustrated in Table 1's top-left (high trust/low distrust) and bottom-right (low trust/high distrust) regions, suggesting that trust and distrust are not completely independent. Meanwhile, the accessibility of the ulterior motive can also lead to distrust in the biased RA. Given the offset in the negative and positive effects of explanations and sponsorship disclosure on distrust, we contend that users' distrust in a biased RA will *not* be significantly affected by the RA's explanations for its organic recommendations, nor by its sponsorship disclosure. Given that these explanations and sponsorship disclosure are not expected to mitigate users' distrust in a biased RA, the mediating effect through perceived PCV and their

interaction effect do not have grounds to exist and are therefore not proposed.

4. Experiment

We conducted a laboratory experiment in the United States with a 3 (RA: neutral, biased with sponsorship disclosure, and biased without sponsorship disclosure) × 2 (explanations: with versus without) between-subjects design (Table 2) to test our hypotheses. We performed a second laboratory experiment in Hong Kong (HK) (see Section 5.5 and Appendix E) to confirm the major results of the first experiment.

4.1. Experimental RA and Treatments

An experimental RA was designed to simulate well-known operational RAs available at leading providers. Aligned with prior studies on RAs (e.g., Wang and Benbasat 2009), this experimental RA was employed to help participants select digital cameras for an experimental task. The experimental RA was validated with the help of several digital camera experts, whose comments were incorporated in the RA design.

The experimental RA first elicited the user's attribute preferences by posing a series of needs-based questions corresponding to different product attributes (e.g., price, brand, and resolution) in the form of user–RA dialogues (Komiak and Benbasat 2006, Wang and Benbasat 2009). The user was provided with three options (levels) for most of the attributes to indicate his/her preferences. Alternatively, the user could indicate that he/she did not have any preference for a particular product attribute. The RA then provided a set of personalized recommendations using different decision strategies based on the manipulations of recommendation neutrality. Figures A.1 and A.2 in Appendix A provide two screenshots of the experimental RA.

4.1.1. Treatment (1)—Neutral RAs vs. Biased RAs. Neutral RAs elicit the user's preferred value of different product attributes and then faithfully integrate these preferences by applying certain decision strategies to generate product recommendations specifically for the user. The neutral RA in this study employed the additive compensatory (AC)[5] strategy, a normative decision strategy, to generate neutral recommendations. The user was provided with a list of 10 recommended products in the order of their fit to the focal user's preferences. The AC-based RA provided neutral recommendations that matched each user's explicit preferences and considered the importance levels of these preferences. This type of RA has been validated as effective in previous studies (Song et al. 2007, Wang and Benbasat 2009).

Table 2. Designs of Experiment

	Explanations for organic recommendations	
RA	With	Without
Neutral	Group 1	Group 2
Biased with sponsorship disclosure	Group 3	Group 4
Biased without sponsorship disclosure	Group 5	Group 6

The recommendation results of the AC-based RA were manipulated for the biased RA. The top three product recommendations were not generated strictly based on the users' attribute preferences, but were always biased toward a specific brand. The Vivitar brand was selected as the least well-known brand of digital camera (see Vivitar cameras presented in Figure A.2 in Appendix A). This brand choice was selected to make the manipulation of recommendation neutrality salient and avoid the possibility that some participants might actually prefer this brand if the RA were biased toward a well-known brand. Our pretest indicates that none of the participants were aware of the Vivitar brand or preferred this brand.

We also manipulated the product attribute levels of the three Vivitar products recommended by the biased RA; they were all dominated by other products in the product database used by the experimental RAs. In other words, the product attributes of the three Vivitar products were less desirable than those of any other products in the database (e.g., a higher price and a lower resolution). Therefore, these Vivitar cameras must be among the least likely recommended products based on users' preferences.

The key difference between the biased and neutral RAs was that the former adjusted the recommendation order that was supposed to be recommended by the latter. The biased RA always recommended the three unknown Vivitar cameras as its top recommendations, which were not likely to be listed as top recommendations by a neutral RA. Except for these three products, the recommended products of the biased RA were also based on the AC strategy (i.e., the same strategy used by the neutral RA) that considered each subject's individual product attribute preferences.

4.1.2. Treatment (2)—Explanations and Sponsorship Disclosure. Explanations were manipulated to include a set of three explanation types—namely, *how* explanations, *why* explanations, and *guidance* (Wang and Benbasat 2007). The biased RA with sponsorship disclosure provides the following disclosure: "Please be advised that the first three recommended products are sponsored products. The sponsor paid for the placements in the recommendation list."

Three buttons (i.e., a "How" button for each *how* explanation, a "Why" button for each *why* explanation, and a "Guidance" button for each *guidance*) were provided by the RAs in the "with explanations" condition for each question in the user–RA dialogue. Wang and Benbasat (2007) rigorously developed and validated these explanation types. An additional button

related to sponsorship disclosure was provided on the product recommendation page in the "biased RA with sponsorship disclosure" condition. Table B.1 in Appendix B provides examples of these explanations.

4.2. Experimental Tasks, Procedures, and Participants

A power analysis for a between-subjects design determined that 240 subjects (40 subjects for each group) can assure a sufficient statistical power of 0.80 to detect an effect size of 0.20, near the medium effect size (f = 0.25) (Cohen 1988). We recruited 247 participants from a large public university in the American Midwest. The participants' ages ranged from 18 to 66 years with a mean age of 23.2 years; 24.7% were older than 24. A total of 48% of the participants were male, and 69.2% had either a full- or part-time job. The participants were surveyed regarding their demographic backgrounds, general web experience, and web-shopping risk attitudes prior to the experimental tasks. No significant difference was observed in gender, age, or employment distribution for participants across the different treatments.

The experimental task was to find a digital camera for the participants themselves with assistance from the experimental RA. The task ended when the participants selected a digital camera and placed it in the shopping cart, but they were asked not to complete the checkout process.

Participation in the experiment was voluntary, and a guaranteed course credit was provided to each participant. Before the experimental tasks, the participants were told that 25% of them would receive an additional reward of $13 based on their performance to motivate them to consider the experiment as a serious online shopping session (Wang and Benbasat 2009). Before the experiment, the participants were informed that they would be asked to justify their product choices and that their performance would be judged based on how convincing their justifications were.

The participants started by answering a demographic questionnaire. They were then asked to watch a video that provided instructions on the experimental task. The video also shows a tutorial RA with the same features as the experimental RA. The video demonstrated the use and navigation of the experimental RA interface. Subsequently, each participant was provided with a hard copy of the instructions and asked to complete the experimental task. The participants were then asked to fill out a form to indicate their product choice and a justification for their choice. Finally, the participants answered the questionnaires with measurement items for the dependent variables and manipulation checks.

4.3. Measurements

The measurement items for trust and distrust were adapted from literature. The measure of trust was based on typical measurements used by Qiu and Benbasat (2009) and Pavlou and Fygenson (2006). The trust items were selected to reflect the users' specific beliefs about the competence, benevolence, and integrity of an RA similar to previous empirical research on trust in information technologies (Vance et al. 2008) and RAs (Wang and Benbasat 2007, 2008). Cho (2006) developed and validated the measurement items for distrust based on the conceptual work of Lewicki et al. (1998). We adapted these items in the current study. We measured PCV by adapting the measurement scale used by Pavlou and Gefen (2005). The measurement items were rated on a seven-point Likert scale ranging from "strongly disagree" (1) through "neutral" (4) to "strongly agree" (7). Tables C.1 and C.2 in Appendix C list all of the measurement items, including those for the manipulation checks. We also measured the following control variables: participants' product knowledge of digital cameras, need for cognition, and RA knowledge.[6]

5. Data Analysis and Results

5.1. Manipulation Checks

We checked the experimental manipulations regarding recommendation neutrality, sponsorship disclosure, and explanations. The participants in the neutral RA group perceived a higher recommendation neutrality than those in the biased RA groups (mean difference = 0.99, $p < 0.001$). Therefore, the manipulation of recommendation neutrality was successful. In H2 and H4, we theorized that perceived PCV is the highest for biased RAs without sponsorship disclosure and the least for neutral RAs, whereas the RAs with sponsorship disclosure fall in-between. We confirmed our theory by running planned contrast analysis tests (Schindler et al. 2011, Uhrich 2011). Results show that the neutral RAs had a lower perceived PCV than biased RAs with sponsorship disclosure (mean difference = −0.76, $p < 0.01$); the perceived PCV for the latter was lower than that for the biased RAs without sponsorship disclosure (mean difference = −0.69, $p < 0.01$).

The perceived transparency of the experimental RA with explanations versus the RA *without* was compared to check the explanation manipulation. A significant difference was observed as follows: RA with explanations (mean = 4.75, SD = 1.44) and RA without explanations (mean = 4.30, SD = 1.25), $t = 2.93$, $p < 0.01$. We also asked the subjects assigned to the experiment groups with explanations whether they considered the RA's "explanations" as improving the RA's effectiveness and helping them understand the RA better (see items in Table C.2 in Appendix C). A one-sample T test comparing the test value of 4.00 (i.e., the scale midpoint) with participants' ratings (mean = 5.18) indicated that explanations improved their understanding about and their utilization of the RAs (difference = 1.18, $p < 0.001$). For *biased RAs*, we also compared the participants' awareness of sponsorship disclosure between the biased RA with sponsorship disclosure versus the RA

Table 3. Descriptive Statistics and Correlations

Variable	Mean	SD	Cronbach's α	1	2	3
1. *Trust*	4.91	1.08	0.92	**0.83**[a]		
2. *Distrust*	3.77	1.17	0.90	−0.66	**0.78**	
3. *Perceived PCV*	4.88	1.46	0.93	−0.45	0.55	**0.96**

[a]Diagonal elements (in bold) are square roots of the average variance extracted (AVE), whereas off-diagonal elements are interconstruct correlations.

without. The following significant difference was determined in the mean of the two items for this manipulation: biased RA *without* sponsorship disclosure (mean = 3.50, SD = 1.54) and biased RA *with* sponsorship disclosure (mean = 5.28, SD = 1.52), $t = 7.51$, $p < 0.001$.

5.2. Measurement Characteristics

Table 3 shows the descriptive statistics of trust and distrust in RAs and perceived PCV. We examined the reliability and validity of these constructs. The reliability indicators measured by Cronbach's α (Table 3) were all above the cited minimum value of 0.7 (Nunnally and Bernstein 1994). The discriminant validity of the constructs was assessed through an exploratory, principal components factor analysis with direct oblimin rotation using SPSS version 22. Table 4 presents the results of the factor analysis.

To assess discriminant validity, Fornell and Larcker (1981) also pointed out that the AVE should be larger than the variance shared between the construct and other constructs in the model (i.e., the squared correlation between two constructs). Table 3 shows that all construct pairs met this requirement.

The discriminant validity was further checked whether the loadings for the items on their targeted construct were higher than those on the other constructs in the model (Chin 1998). Gefen and Straub (2005) provided a restrictive guideline stating that the minimum difference between item loadings and cross-loadings should be larger than 0.10 to establish discriminant validity besides the aforementioned AVE analysis. This restrictive guideline has been widely adopted in the information systems literature to assess discriminant validity (Gefen and Straub 2005, Zhang and Sun 2009, Xu et al. 2013). Table 4 contains the loadings and cross-loadings for the items used in this study. All items were loaded higher on their constructs than they were on any other constructs. Furthermore, the differences in all cases were more than 0.10, with most of them being larger than 0.20. These results confirm that the constructs are factorially distinct from each other, and that all items used to operationalize a particular construct were loaded to a single factor.

Table 4. Factor Analysis Results

Construct	Item	Factor 1	Factor 2	Factor 3
Trust in RAs	Trust1	**0.778**	−0.059	0.217
	Trust2	**0.837**	−0.197	0.108
	Trust3	**0.813**	−0.175	0.091
	Trust4	**0.654**	−0.289	0.395
	Trust5	**0.776**	−0.297	0.185
	Trust6	**0.736**	−0.309	0.209
	Trust7	**0.753**	−0.392	0.125
Distrust in RAs	Distrust1	−0.138	**0.770**	−0.193
	Distrust2	−0.072	**0.814**	−0.119
	Distrust4	−0.146	**0.831**	−0.211
	Distrust6	−0.480	**0.651**	−0.032
	Distrust7	−0.468	**0.695**	0.031
	Distrust8	−0.553	**0.631**	−0.034
	Distrust9	−0.499	**0.664**	−0.027
Perceived PCV	PCV1	−0.294	0.154	**0.710**
	PCV2	−0.118	0.090	**0.810**

Note. Numbers in bold are factor loadings with the strongest association to the corresponding underlying latent variable.

5.3. Further Confirmation of the Coexistence of Trust and Distrust as Distinct Constructs

We further assessed the coexistence of trust and distrust in addition to the reported evidence of the discriminant validity of these two constructs. We checked the distribution of the participants' different levels of trust versus distrust of the experimental RA using a middle split (4.0/7.0) of their self-reported trust and distrust scores. As suggested by Lewicki et al. (1998), we categorized the participants into four combinations of trust and distrust levels as follows: high trust/low distrust, low trust/high distrust, high trust/high distrust, and low trust/low distrust (the results are presented in Table D.1 in Appendix D). We determined that high trust/low distrust (49.8%) and high trust/high distrust (26%) were more prevalent, followed by low trust/high distrust (22%) and low trust/low distrust (2.2%). The existence of all of these categories confirms the coexistence of trust and distrust as two distinct constructs.

5.4. Hypothesis Testing Results

To test H1A, H1B, H2A, and H2B, we conducted a MANCOVA with trust and distrust as the dependent variables to examine the overall difference between a biased RA with sponsorship disclosure and a neutral RA. The participants' self-reported product knowledge of digital cameras, need for cognition, and RA knowledge were included as control variables in the MANCOVAs. Results show that the treatment effects of recommendation neutrality were significant ($p < 0.001$). Therefore, we conducted two separate ANCOVAs for trust and distrust. The ANCOVA results reported in Table 5 indicate that the neutral RA led to a higher level of trust compared with the biased RA with sponsorship disclosure (mean difference = 0.56, $p < 0.01$), thereby supporting H1A. The neutral RA generated lower distrust than the biased RA with sponsorship disclosure (mean difference = −0.68, $p < 0.01$), thereby supporting H1B.

Table 5. Results of ANCOVAs: Effects of Recommendation Neutrality and Explanations (Groups 1–4)

Source	Type III sum of squares	d.f.	Mean square	F	p-value
Dep. var.: *Trust in RAs*					
Need for cognition	2.09	1	2.09	2.13	0.15
Product knowledge	1.29	1	1.29	2.32	0.25
RA knowledge	4.78	1	4.78	4.89	0.03
Recommendation neutrality	9.86	1	9.86	10.07	<0.01
Explanations	7.59	1	7.59	7.75	<0.01
*Neutrality * Explanations*	3.88	1	3.88	3.96	0.04
Dep. var.: *Distrust of RAs*					
Need for cognition	2.060	1	2.060	1.767	0.18
Product knowledge	0.434	1	0.434	0.373	0.54
RA knowledge	4.897	1	4.897	4.200	0.04
Recommendation neutrality	15.313	1	15.313	13.133	<0.01
Explanations	0.825	1	0.825	0.707	0.40
*Neutrality * Explanations*	2.540	1	2.540	2.179	0.14

We then conducted the bootstrapping mediation test (Preacher and Hayes 2008a, b) and the Sobel test (Sobel 1982) to examine the mediating role of perceived PCV (i.e., 2A/2B and H4).[7] Preacher and Hayes (2008a, b) determined that the key condition is to test whether the indirect effect is significant.[8] Accordingly, this study applied PROCESS for SPSS v2.13 (Hayes 2012, 2013) with 1,000 bootstrapped samples to estimate such an indirect effect ($a \times b$).

Regarding the mediating effects of perceived PCV for the comparison of a biased RA with sponsorship disclosure and a neutral RA (i.e., H2A and H2B), the results show that the significant indirect effect of recommendation neutrality on trust was 0.45, which had a 95% confidence interval that did not include zero [0.11, 0.61]. Similarly, the indirect effect of recommendation neutrality on distrust was significant with an indirect effect of −0.27, which had a 95% confidence interval that did not include zero [−0.42, −0.16]. The Sobel test also indicated a significant indirect effect of recommendation neutrality on trust and distrust through PCV ($p < 0.001$). The direct effect of recommendation neutrality on trust was not significant ($p = 0.14$); hence, perceived PCV fully mediated its effects on *trust*, which supports H2A. However, the direct effect of recommendation neutrality on distrust was significant ($p < 0.01$). Therefore, PCV partially mediated the relationship between recommendation neutrality and *distrust*, partially supporting H2B.

We tested H3A, H3B, H4, and H5 by focusing on the *biased RA* conditions (groups 3–6 in Table 2). Two ANCOVAs were conducted: one with trust as the dependent variable and the other with distrust as the dependent variable. The participants' self-reported product knowledge of digital cameras, need for cognition, and RA knowledge were included as control variables in the ANCOVAs.

Table 6. Results for H3 and H5 (Biased RA Conditions Only)

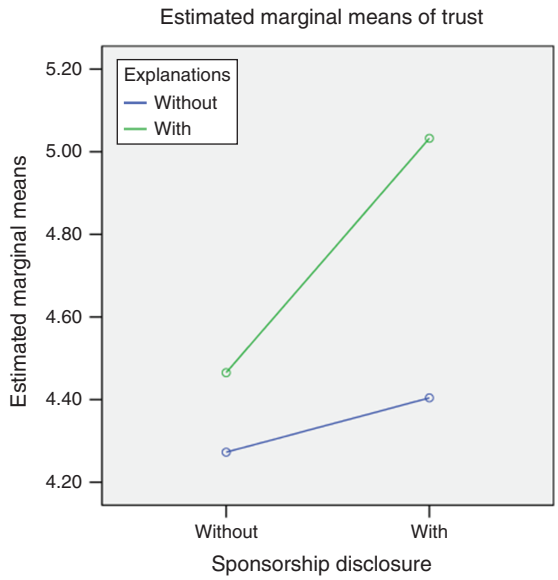
Source	Type III sum of squares	d.f.	Mean square	F	p-value
Dep. var.: *Trust in RA*					
Need for cognition	3.962	1	3.962	3.185	0.07
Product knowledge	2.007	1	2.007	1.613	0.20
RA knowledge	1.595	1	1.595	1.138	0.28
Sponsorship disclosure	5.353	1	5.353	4.302	0.04
Explanations	7.039	1	7.039	5.657	0.01
*Sponsorship disclosure * Explanations*	5.095	1	5.095	4.095	0.04
Dep. var.: *Distrust in RAs*					
Need for cognition	2.040	1	2.040	1.359	0.24
Product knowledge	0.024	1	0.024	0.016	0.90
RA knowledge	0.889	1	0.889	0.619	0.40
Sponsorship disclosure	0.417	1	0.417	0.278	0.59
Explanations	2.640	1	2.640	1.758	0.18
*Sponsorship disclosure * Explanations*	1.013	1	1.013	0.675	0.41

Table 6 reports the ANCOVA results, which show the significant main effects of explanations ($p < 0.05$) and sponsorship disclosure ($p < 0.05$), as well as their significant interaction effects on trust in biased RAs ($p < 0.05$). Thus, H3A, H3B, and H5 were all supported. However, none of these three effects on *distrust* was significant ($p > 0.1$), indicating that RA's explanations and sponsorship disclosure did not affect users' distrust in a biased RA.

Given that an interaction effect was significant for trust, we conducted contrast analysis tests (Schindler et al. 2011) to examine when trust in biased RAs can be best improved. Figure 2 demonstrates that the presence of both explanations and sponsorship disclosure was significantly better than explanations only ($p < 0.05$), sponsorship disclosure only ($p < 0.05$), and the absence of explanations and sponsorship disclosure ($p < 0.01$). No significant differences were found among the latter three. Therefore, the combination of explanations and sponsorship disclosure was considered to lead to the highest trust level in a biased RA. In other words, only when both explanations and sponsorship disclosure are provided can users' trust in a biased RA significantly increase.

H4's testing results show that the indirect effect of sponsorship disclosure on trust was 0.19 ($p < 0.05$) with a 95% confidence interval between [0.03, 0.41]. The direct effect of sponsorship disclosure on trust was insignificant ($p = 0.31$). Thus, PCV fully mediated the effects of sponsorship disclosure on trust, and H4 was supported. Furthermore, we examined whether the interaction effect of sponsorship disclosure and explanations on trust in biased RAs was mediated by PCV. Preacher et al. (2007) and Hayes (2012, 2013) outlined that the procedure to test whether a moderation effect is mediated is to estimate the indirect effect of the interaction on dependent variable Y through the mediator M. Based on mathematics modeling, this indirect

Figure 2. (Color online) For Biased RAs: Interaction Between Sponsorship Disclosure and Explanations



effect is actually the difference between the total effect of the interaction effect on Y when M is absent and the direct effect of the interaction effect on Y when M is present. The idea is the same as in the pure mediation test where the indirect effect is the difference between the total and direct effects (see Endnote 8). Hayes' mediated moderation test guideline supported by PROCESS macro has been recently adopted in the fields of psychology (Sullivan et al. 2010), marketing (Cavanaugh 2014, Humphreys and Latour 2013, Whelan and Dawar 2016), and information systems (Blohm et al. 2016). Applying this approach reveals a significant indirect effect ($\beta = 0.34$, 95% CI = 0.02, 0.78) of the interaction, supporting that the interaction effect between the RA's sponsorship disclosure and explanations on trust was mediated by perceived PCV. Similar results were obtained by using the Sobel test.

5.5. Discussion

Our experimental results offer compelling evidence in support of most of the hypotheses. Users' trust/distrust in a neutral RA is higher/lower than their trust/distrust in a biased RA with sponsorship disclosure (H1A/1B). Sponsorship disclosure increases users' trust in a biased RA (H3B) but cannot lower their distrust. Moreover, explanations play a critical moderating role in the effect of sponsorship disclosure on trust (H5) but *not* for distrust in a biased RA. Perceived PCV fully mediates the effects of recommendation neutrality on trust (H2A) and those of sponsorship disclosure on trust (H4), and partially mediates the effects of recommendation neutrality on distrust (H2B). A *biased* RA must provide both explanations for its organic recommendations and sponsorship disclosure to obtain user trust in the biased RA (H5).

We conducted a second laboratory experiment with a 2 (recommendation neutrality: biased versus neutral) × 2 (explanations and sponsorship disclosure: with versus without) between-subjects design (Table E.1 in Appendix E) in HK. This HK experiment attempted to demonstrate the robustness of the key results from the U.S. experiment (i.e., the effect of recommendation neutrality on trust and distrust, and the positive effect of biased RA's sponsorship disclosure and explanations in tandem on trust but not distrust). The HK experiment differed from the U.S. one in several aspects. First, the HK experiment recruited subjects from HK to enhance the result robustness. Second, we checked the perceived anomalies in product recommendations as a proxy for perceived PCV.[9] We also asked the participants an open-ended question to have them elaborate on their perceived PCV of the experimental RA. Third, we did not separate explanations and sponsorship disclosure for biased RAs in the HK experiment because the U.S. experiment results show that the combination of explanations and sponsorship disclosure matters.

Appendix E shows that the HK experiment results provide compelling evidence that recommendation neutrality significantly influences users' trust and distrust. The results also confirm that explanations and sponsorship disclosure provided by a biased RA can collectively mitigate the negative effect of the biased RA (versus neutral RAs) on trust but cannot lower the effect on distrust. Hence, the U.S. experiment findings can be generalized into a sample in a different geographic location.

6. General Discussions

6.1. Theoretical Contributions

This study provides significant theoretical contributions. First, this study contributes to the understanding of the recommendation neutrality of RAs by applying the PCV perspective. Existing research on RAs primarily focuses on their beneficial effects with an assumption that these technologies intend to provide neutral advice for users (Wang and Benbasat 2009). Recent studies examine the online sponsorship phenomenon, but they mainly focus on the effectiveness of such sponsorships from the sponsor's perspective (e.g., the number of increased visits to the advertised website) (Ghose and Yang 2009, Rutz et al. 2011). This study theorizes and empirically confirms that biased RAs without sponsorship disclosure lead to the highest perceived PCV level, followed by biased RAs with sponsorship disclosure. The neutral RAs have the lowest perceived PCV level. More importantly, the effects of recommendation neutrality and sponsorship disclosure on trust are fully mediated by the perceived PCV. When biased RAs disclose the sponsorships involved, users may have to adjust their expectations to some

extent regarding what they can receive from the RAs and accordingly revise their psychological contract with RAs, although users cannot completely "forgive" biased RAs.

Second, our study contributes to the understanding of different patterns in trust improvement versus potential distrust mitigation of RAs. The separability of the positive and negative evaluative processes has been extensively discussed in the fields of psychology (Cacioppo and Berntson 1994, Cacioppo et al. 1997), psychobiology (Martin and Levey 1978), management (Lewicki et al. 1998), and information systems (Dimoka 2010). However, limited research has been conducted to investigate the differences between trust and distrust with respect to their improvement (trust) or decrease (distrust). This study fills this gap. Our findings attest that trust emerging from the understanding of the trustee can be robust (Lee and See 2004). The explanations for organic recommendations provided by an RA can help improve users' trust in the RA even when that RA is biased toward sponsors, as long as the biased RA discloses the sponsorships involved and provides these explanations. By contrast, *distrust* is difficult to resolve once formed because it is based on goal and value incongruence (Connelly et al. 2012, Sitkin and Roth 1993). Sponsorship disclosure or explanations provided by an RA cannot remove this incongruence to mitigate users' distrust. Our theorizing of the different roles of explanations in moderating the effects of sponsorship disclosure on trust versus distrust advances our understanding of users' different evaluative spaces for trust versus distrust.

Third, it contributes to existing literature by investigating the effects of sponsorship disclosure provided by a biased RA. We tackled this issue by adopting a contingency approach to investigate the theoretical boundaries, in which sponsorship disclosure can or cannot affect users' trust in biased RAs. In particular, we examined whether the explanations amplify the positive effects of a biased RA's sponsorship disclosure on trust. Wang and Benbasat (2007) empirically attested the *main* effects of explanations on users' trust in a neutral RA. Nevertheless, the RA technologies examined in previous studies are *neutral* and designed to find the most suitable products that solely match users' preferences (e.g., Komiak and Benbasat 2006). Recommendation neutrality is not assumed in this study. The biased RA attempts to promote certain products that may not be the most suitable for the user. Our results attested to the important moderating role of explanations for organic recommendations provided by a biased RA. The sponsorship disclosure of a biased RA increases users' trust only when these explanations are also provided. Thus, a biased RA that provides either only explanations or only sponsorship disclosure is insufficient to improve user trust; both are required.

6.2. Practical Implications

This study has significant implications for practitioners. First, given the mediating role of perceived PCV in the relationships examined in this study, RA providers need to manage users' psychological contracts and deliver product recommendations that are consistent with users' expectations to lower the potential perceived PCV of users. Specifically, we advise practitioners to explain their need to recommend sponsored products, such as obtaining necessary revenues for business survival and better serving RA users. This approach can allow users to understand the perspective of providers of biased RAs and change their beliefs that neutral recommendations are the norm. Subsequently, users will be less likely to perceive PCV when receiving biased product recommendations.

Second, RA providers should cautiously apply designs that can induce users' distrust in the RAs. The potential for the coexistence of trust and distrust is considerable in online environments (Häubl and Murray 2006, Wang et al. 2014, Winder 2011). Our results indicate that the factors that lead to distrust are not always the same as those that influence trust. Furthermore, dealing with users' distrust is more difficult than improving their trust. The technology provider can be compelled to directly address the negative expectations of users regarding the injurious conduct of RAs. Alternatively, the RA provider can allow users to decide whether to receive sponsored products to ease their concerns. Further research should be conducted to fully determine the effectiveness of these approaches.

Third, RAs should provide explanations (and sponsorship disclosure in the case of biased RAs) to faithfully disclose their inner workings to ensure users' trust. Prior studies present a similar suggestion (Wang and Benbasat 2007), but these studies primarily focus on neutral technologies. In practice, many providers of biased RAs do not disclose clear sponsorships to users probably because of their worries about users' negative responses once the users are informed about the use of sponsorships. Our results show that users' trust in a biased RA can be maintained if the biased RA offers both explanations and sponsorship disclosure.

6.3. Limitations and Future Research

This study has several limitations. First, this study examines a situation that biased recommendations are perceptible to users. In other words, the manipulations of biased recommendation were made salient with products that do not fit users' product preferences in the experiment. However, biases in the recommendation results can be made subtler. For example, the RA can be biased toward brands that fit users' preferences, but users may be unable to identify such biases during their first interaction with the RA (Jiang et al. 2016). In this case, a user may not perceive PCV resulting from the biased recommendations, and his/her

trust and distrust in the RA will not be affected by the biased recommendations. Although social media and monitoring websites such as propublica.org can spread such subtle biases rapidly and raise consumers' awareness, readers should be cautious about generalizing our results to such a context with subtly biased products that fulfill users' preferences. Future research should consider whether users can easily recognize these subtle biases and their impacts. Second, we did not look into RA designs that can address users' distrust, although explanations and sponsorship disclosure can be applied to acquire users' trust in a biased RA. We also found that perceived PCV only partially mediates the effects of recommendation neutrality on *distrust*, which opens up future research opportunities to examine what other mechanisms can explain the effects of recommendation neutrality on distrust. Third, this study used self-reported measures for trust and distrust, instead of using emerging techniques, such as a neurophysiological method (Dimoka 2010). We recommend that future research explore such approach because additional insights can be generated. Finally, future research should examine whether trust and distrust have different effects on certain outcome variables, such as actual purchase behavior, and word-of-mouth behavior. Another possible extension is to examine the effect of explanations on the perceived value and purchase intention of the sponsored product (i.e., the top recommended products in our experiment).

6.4. Conclusions

Online search and recommendation neutrality has become a hot topic for researchers, practitioners, and policy makers. Consumer ambivalence about recommendation services is common. A biased RA lowers users' trust and increases their distrust. However, users' trust in a biased RA could be increased when the RA provides both sponsorship disclosure and explanations to faithfully disclose its inner workings. By contrast, the effects of lack of recommendation neutrality on *distrust* cannot be resolved easily by these explanations. Our findings extend the understanding of the effects of sponsorship disclosure and recommendation neutrality on trust and distrust from the PCV perspective, and of the moderating role of explanations in influencing the effects of sponsorship disclosure on trust in a biased RA. We provide guidelines for practitioners to offer effective RAs that can satisfactorily serve online consumers.

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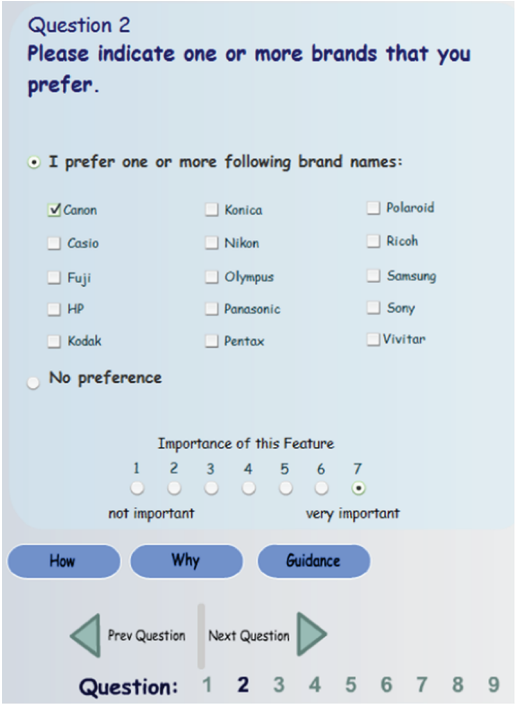
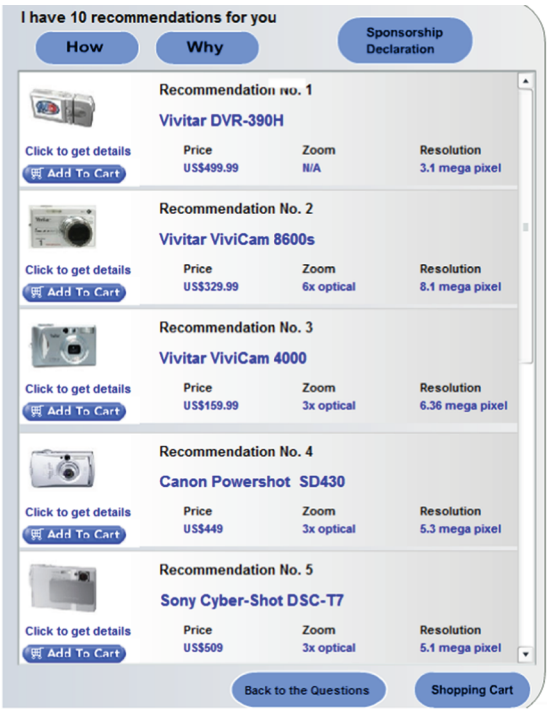
Appendix A. Screenshots of the Experimental RA

Figure A.1. (Color online) User–RA Dialogue



A sample question (brand preferences) from the RA with explanations

Figure A.2. (Color online) Product Recommendations from the RA



Biased RA (vivitar cameras recommended) with explanations and sponsorship disclosure

Appendix B

Table B.1. Examples of the Explanations and Sponsorship Disclosure in the Experimental RA

A question in the user–system dialogue	How will you view or print the pictures you have taken? (1) Most likely, I will view them on computers or print them in sizes of 4 × 6 or 5 × 7. (2) Probably I will print some of them in sizes of 8 × 10 or 11 × 14 or even 13 × 19. (3) I may print some of them in large sizes of at least 16 × 20. (4) No preference.
How explanation	Your decision about whether or not to print the pictures and the sizes of the prints will determine the appropriate resolution level of a digital camera. Cameras with your desired resolution level will be given higher priority in my recommendations. Digital cameras capture images as pixel elements, known as pixels, and a megapixel is equal to one million pixels. Digital images are made up of thousands of these tiny, tile-like picture elements. The more pixels, the higher the image resolution. Specifically, the four options will determine the following resolution levels: (1) Below 6 megapixels. (2) Between 6 megapixels and 9 megapixels. (3) Higher than 9 megapixels. (4) All cameras will be considered.
Why explanation	Different ways to deal with pictures are very important for digital camera users. Photos can be kept in electronic format (e.g., in a computer). You can then browse them, email them, or put them on web pages to share with others worldwide. For many users, it is also very important that their photos can be printed in various sizes. For example, for framing (e.g., a wall frame or a table frame), you may need larger prints.
Guidance	Nearly all cameras are capable of printing small-sized photos such as 4 × 6 or 5 × 7, and those cameras with a resolution of 3 M to 6 M are used to do this job. Cameras are now designed to meet the need of customers to print high-quality pictures of a larger size. Thus, most cameras are capable of printing medium-sized photos (11 × 14), while half of the cameras can print large-sized photos (13 × 19). Of course, cameras for larger prints are generally much more expensive, and the feature of being able to generate very large prints will limit your options. Therefore, be careful not to over-estimate your needs.
Sponsorship disclosure	Please be advised that the first three recommended products are sponsored products. The sponsor paid for the placements in the recommendation list.

Appendix C. Measurements

Table C.1. Measurement Items For Trust, Distrust, and PCV

Items	Wording	U.S. experiment	HK experiment
	Trust		
Trust1	This recommendation system was competent in recommending digital cameras.[a]	Yes	Yes
Trust2	This recommendation system performed its role of recommending digital cameras effectively.	Yes	Yes
Trust3	Overall, this recommendation system was capable of providing suitable digital camera recommendations.	Yes	Yes
Trust4	This recommendation system's dealings with me were in my best interest.	Yes	Yes
Trust5	This recommendation system's dealings with me felt like that it would do its best to help me.	Yes	Yes
Trust6	This recommendation system's recommendations to me were truthful.	Yes	Yes
Trust7	I would characterize this recommendation system's dealing with me as honest.	Yes	Yes
	Distrust		
Distrust1	This recommendation system was designed to exploit customers' vulnerability given the chance.	Yes	Yes
Distrust2	This recommendation system was designed to engage in harmful behavior to customers to pursue its own interest.	Yes	Yes
Distrust3	This recommendation system was designed to operate in an irresponsible manner.	No	Yes

Table C.1. (Continued)

Items	Wording	U.S. experiment	HK experiment
	Distrust (continued)		
Distrust4	This recommendation system was designed to perform the business with customers in a deceptive way.	Yes	Yes
Distrust5	This recommendation system's recommendations to me were fraudulent.	No	Yes
Distrust6	This recommendation system is capable of engaging in harmful behavior by recommending biased cameras.	Yes	No
Distrust7	This recommendation system has the ability to maliciously manipulate the cameras recommended.	Yes	No
Distrust8	This recommendation system is capable of deceiving users by recommending biased cameras.	Yes	No
Distrust9	I suspect that this recommendation system is interested in just its own well-being, not mine.	Yes	No
	Perceived Psychological Contract Violations (*PCV*)		
PCV1	In general, this recommendation system failed to meet its obligations to me during our interactions.	Yes	Not measured
PCV2	In general, this recommendation system did a poor job of meeting its obligations to me during our interactions.	Yes	
PCV3	In general, this recommendation system fulfilled the most important obligations to me during our interactions. (*reverse, dropped*)	Yes	

Notes. The U.S. experiment was conducted subsequent to the HK experiment with additional four measurement items related to competence belief of distrust. Two items from the HK experiment were dropped in the U.S. one as their loadings were lower than 0.55.

[a]We used the term "virtual advisor," instead of "recommendation system," for the experimental RA in the supplementary experiment conducted in Hong Kong.

Table C.2. Items For Manipulation Checks

Manipulations	Check items	Sources
Recommendation neutrality	This recommendation system appeared to be biased toward certain camera vendor(s).	Darke and Ritchie (2007), Poortinga and Pidgeon (2003)
	This recommendation system provided misleading recommendations.	
	When giving recommendations, this recommendation system distorted factors in favor of certain camera vendor(s).	
Explanations (for all subjects)	This recommendation system made its reasoning process clear to me.	Wang (2005)
	I could easily understand this recommendation system's reasoning process.	
	This recommendation system's logic in providing advice was clear to me.	
Explanations (for subjects assigned to the experimental groups with explanations treatment only)	Viewing the explanations provided by this recommendation system can improve my utilization of the recommendation system.	Wang (2005)
	Viewing the explanations provided by this recommendation system can help me understand this recommendation system.	
	Viewing the explanations provided by this recommendation system can increase the effectiveness of using this recommendation system.	
Sponsorship disclosure	This recommendation system informed me of the sponsored products.	Developed for this study
	This recommendation system made the sponsorship of the recommended products explicit to me.	

Appendix D

Table D.1. Counts (Percentage) of Participants with Different Levels of Trust and Distrust

	Experiment in United States			Experiment in HK		
	Low distrust	High distrust	Total	Low distrust	High distrust	Total
High trust	115 (49.8%)	60 (26%)	175 (75.8%)	74 (45.1%)	10 (6.1%)	84 (51.2%)
Low trust	5 (2.2%)	51 (22%)	56 (24.2%)	26 (15.9%)	54 (32.9%)	80 (48.8%)
Total	120 (51.9%)	111 (48.1%)	231 (100%)	100 (61.0%)	64 (39.0%)	164 (100%)

Notes. In the U.S. experiment, 16 participants indicating "neutral" (4.00/7/00) to either trust or distrust were removed. In the HK experiment, 35 of these participants were removed.

Appendix E. Supplementary Experiment Conducted in Hong Kong

Table E.1 illustrates the designs of the HK experiment. The general designs and differences between this experiment and that of the U.S. experiment are presented in Section 5.5.

E.1. Experimental Tasks, Procedures, and Participants

A total of 200 participants were recruited from a public university in HK. One participant encountered network problems during the experiment and was removed from the data set. The sample size of 199 (50 subjects for each group) can assure a sufficient statistical power of 0.80 to detect a small-to-medium effect size ($f = 0.15$ to 0.25) (Cohen 1988). This sample comprised 60 males and 139 females with an average age of above 21 years. No significant differences were observed among the participants assigned to different experimental groups with respect to different dependent variables of interest to this research. The incentive, stimuli, and procedure were similar to those in the main experiment conducted in the United States.

E.2. Measurements, Manipulation Checks, and Construct Validity

We conducted similar manipulation checks as in the U.S. experiment. First, the manipulation of recommendation neutrality was successful given the following significant difference in perceived recommendation neutrality: neutral RA (mean = 4.33, SD = 1.12) and biased RA (mean = 3.12, SD = 1.18), $t = 7.41$, $p < 0.001$. We also checked the perceived anomalies in product recommendations as a proxy for the PCV. We asked the participants an open-ended question to have them elaborate on their perceived PCV of the experimental RA. We found that the participants primarily focused on the RA outcomes (i.e., product recommendations) and considered the anomalies in the product recommendations as the main PCV form. A significant difference was noted in the perceived anomalies between the neutral and biased RAs: neutral RA (mean= 4.04, SD = 1.16) and biased RA (mean = 5.06, SD = 1.25), $t = 5.97$, $p < 0.001$. This result provided evidence of the validity of using the PCV perspective in this study.

Second, the explanation manipulation, including sponsorship disclosure if applicable, was successful after checking the perceived RA transparency. Specifically, the following significant difference was found: RA with explanations/sponsorship disclosure (mean = 5.08, SD = 0.99) and RA without explanations/sponsorship disclosure (mean = 3.98, SD = 1.43), $t = 6.30$, $p < 0.001$. For the *biased RAs*, we compared the participants' awareness of sponsorship disclosure between the RA with sponsorship disclosure versus the RA without. The following significant difference was identified: biased RA *without* sponsorship disclosure (mean = 3.03, SD = 1.33) and biased RA *with* sponsorship disclosure (mean = 4.74, SD = 1.23), $t = 6.53$, $p < 0.001$. This significant difference provided evidence of the successful manipulations of sponsorship disclosure. Table E.2 shows the mean and standard deviation for the four experimental groups in terms of trust and distrust.

We examined the reliability and validity of trust and distrust. Table E.3 shows the results of an exploratory principal component factor analysis. We applied the same analyses as those for the U.S. experiment and found that the data demonstrated good convergent and discriminant validity.

Table E.1. Designs of HK Experiment

	Explanations and sponsorship disclosure	
	With	Without
Recommendation neutrality		
Neutral RA	Group 1	Group 2
Biased RA	Group 3	Group 4

E.3. Further Confirmation of the Coexistence of Trust and Distrust as Distinct Constructs

Similar to the U.S. experiment, we categorized the participants into four combinations of trust and distrust levels and confirmed the existence of these four combinations (see Table D.1 in Appendix D). We found that high trust/low distrust (45.1%) and low trust/high distrust (32.9%) were more prevalent, followed by low trust/low distrust (15.9%) and finally high trust/high distrust (6.1%). This result verifies the coexistence of trust and distrust as two distinct constructs.

E.4. ANCOVA Results

We conducted two ANCOVAs, one with trust as the dependent variable and the other with distrust as the dependent variable. The participants' self-reported product knowledge of digital cameras was included as a control variable. Table E.4 shows the ANCOVA results. The main effects of recommendation neutrality on trust and distrust were both significant ($p < 0.001$). The participants' trust in the biased RA was significantly lower than their trust in the neutral RA, whereas their distrust in the biased RA was significantly

Table E.2. Mean (Standard Deviation) of Variables and Correlations (HK Experiment)

	Neutral		Biased			
	Explanations and sponsorship disclosure		Explanations and sponsorship disclosure		Correlation	
	With	Without	With	Without	1	2
1. Trust	4.28 (0.89)	4.25 (1.12)	4.09 (1.08)	3.29 (1.05)	0.88[a]	
2. Distrust	3.52 (0.96)	3.46 (0.91)	3.96 (0.90)	4.29 (0.85)	−0.70	0.80

[a]Diagonal elements are square roots of the AVE, and the off-diagonal element is interconstruct correlation.

Table E.3. Reliability and Factor Analysis (HK Experiment)

Construct	Cronbach's α	Item	Factor 1	Factor 2
Trust in RAs	0.92	Trust1	**0.89**	0.04
		Trust2	**0.86**	−0.003
		Trust3	**0.87**	−0.10
		Trust4	**0.56**	−0.32
		Trust5	**0.69**	−0.09
		Trust6	**0.62**	−0.29
		Trust7	**0.60**	−0.25
Distrust of RAs	0.86	Distrust1	0.03	**0.63**
		Distrust2	0.12	**0.91**
		Distrust3	−0.16	**0.64**
		Distrust4	−0.20	**0.73**
		Distrust5	−0.23	**0.76**

Note. Numbers in bold are factor loadings with the strongest association to the corresponding underlying latent variable.

higher than their distrust in the neutral RA. These results were generally consistent with those found in H1A/1B.

The interaction effect on trust between recommendation neutrality and explanations was significant ($p < 0.05$). However, this effect was insignificant for *distrust* ($p > 0.05$). Specifically, when explanations and sponsorship disclosure were *not* provided by the biased RA, recommendation neutrality significantly affected trust in an RA. By contrast, no significant effect was found between the neutral and biased RAs when explanations and sponsorship disclosure were provided. Hence, the effect of recommendation neutrality on *trust* was larger when explanations and sponsorship disclosure were *not* provided than when they were provided. This result is generally congruent with the key findings in H5 (i.e., the provision of both explanations and sponsorship disclosure can effectively improve trust in the biased RAs).

Table E.4. ANCOVA Results (HK Experiment)

Source	Type III sum of squares	d.f.	Mean square	F	p-value
Dep. var.: *Trust in RAs*					
Recommendation neutrality	15.98	1	15.98	14.91	<0.001
Explanations	8.81	1	8.81	8.22	0.005
Product knowledge	1.98	1	1.98	1.85	0.18
Neutrality ∗ Explanations	5.98	1	5.98	5.58	0.02
Error	207.90	194	1.07		
Dep. var.: *Distrust of RAs*					
Recommendation neutrality	21.02	1	21.02	26.18	<0.001
Explanations	0.40	1	0.40	0.50	0.48
Product knowledge	0.42	1	0.42	0.53	0.47
Neutrality ∗ Explanations	2.16	1	2.16	2.69	0.10
Error	155.71	194	0.80		

E.5. Discussion

This HK experiment provides compelling evidence that the key findings of the U.S. experiment can be generalized to a sample at a different geographic location. Users' trust/distrust in a neutral RA is higher/lower than their trust/distrust in a biased RA. Explanations and sponsorship disclosure can collectively mitigate the negative effect of the lack of neutrality on trust, but distrust in a biased RA cannot be resolved even when both explanations and sponsorship disclosure are provided.

Endnotes

[1] http://www.searchengineland.com/a-letter-to-the-ftc-regarding-search-engine-disclosure-124169.

[2] This method is different from a collaborative-filtering method, which is based on the opinions of like-minded people to generate recommendations for users (Xiao and Benbasat 2007), such as the method used by Netflix.

[3] Even for these types of subtle biases, we believe that users can detect them sooner or later through repeated usage or learning from other users' experiences. News and word-of-mouth travel fast in the Internet age. Aside from traditional consumer reports, some monitoring websites (e.g., propublica.org) have reported subtle biases, such as the following: (1) Orbitz showed higher-priced hotels to owners of Mac computers, and (2) Staples offered the same products at higher prices to people living in a certain ZIP code (https://www.propublica.org/article/breaking-the-black-box-when-algorithms-decide-what-you-pay).

[4] http://www.searchengineland.com/a-letter-to-the-ftc-regarding-search-engine-disclosure-124169.

[5] Additive compensatory is typically assumed and used in conjoint analysis (Green and Srinivasan 1990, Klein 1987, Schaupp and Bélanger 2005).

[6] The measures for product knowledge were adapted from Xu et al. (2011). One sample item is, "I am familiar with basic camera specifications (e.g., resolution and LCD display)." The measures for need for cognition were adapted from Tam and Ho (2005). One sample item is, "I prefer to do something that challenges my thinking abilities rather than something that requires minimal thought." The single item measure for RA knowledge was adapted from Bhattacherjee and Sanford (2006). The item is, "How knowledgeable are you in using the following web-based technologies: online recommendation systems?"

[7] We examined the potential threat of common method biases (CMB) given that both PCV and trust/distrust were measured through self-reported measurements. We generally followed the design and statistical control methods of Podsakoff et al. (2003) to deal with the CMB,

including the marker-variable technique suggested by Malhotra et al. (2006) and Pavlou et al. (2007). Our results show that the CMB was not a major concern in the current study.

[8] When X (independent variable) has an effect on Y (dependent variable), the path c quantifies this effect, which is referred to as the total effect of X on Y. When the effect of X on Y is mediated by M (the mediator), the effect of X on M can be represented by the path a, whereas the effect of M on Y controlling for X can be represented by the path b, and c' represents the effect of X on Y controlling for M. In this mediation situation, X has an indirect effect on Y through M and a direct effect on Y. The total effect of X on Y is equal to the sum of the direct and indirect effects, that is, $c = c' + a \times b$. Accordingly, the indirect effect is the difference between the total effect and direct effect of X on Y: $a \times b = c - c'$. When $a \times b$ (indirect effect) is significantly different from a value of zero, it means that a significant difference between c and c' exists. That is, the effect of X on Y decreases substantially when M is entered simultaneously with X as a predictor of Y, a key criterion for the mediation to occur (Baron and Kenny 1986, Preacher and Hayes 2008a).

[9] Three items on a seven-point Likert scale were used to measure perceived anomalies: "I received unexpected product recommendations provided by this virtual advisor," "I noticed anomalies in the product recommendations provided by this virtual advisor," and "I noticed inconsistencies between my requirements and the product recommendations provided by this virtual advisor" (Johnson et al. 1993).

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